Sensing and Augmenting for Adaptive Assembly Strategies

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This paper introduces an innovative integrated approach to design, simulate, and test assembly strategies that adapt to tolerances and geometric variations. Traditional Design for Assembly (DfA) methods, while effective for well-known materials and predetermined design parameters, exhibit limitations in addressing unknown parameters and unpredictable material behavior. Our research employs techniques such as 3D scanning, laser projection, and early 1:1 prototyping to develop a flexible adaptive assembly model, accommodating last-minute structural and site condition changes.
We demonstrate our novel workflow through the assembly and installation of a large-scale demonstrator at the AEDES gallery in Berlin, consisting of 24 variable-size 3D printed panels attached to a brick wall. By creating an initial assembly model using early detailing and a preliminary assembly sequence, we maximized the number of adaptable design parameters. An agent-based model was utilized to identify wall connection locations based on site constraints and fastener accessibility. Upon completing panel production, we used industrial laser projectors to compare digital files to actual pieces and 3D scanning to acquire accurate panel connection positions. This information enabled the regeneration of connection points with precise angles and dimensions and informed the final assembly sequence. On-site laser projection facilitated the efficient assembly of the structure. Our approach paves the way for more accurate and adaptable construction methods in complex architectural projects.

Keywords: *Assembly Information Modeling, 3D Scanning, Laser Projection, Design for Assembly, Material Behavior.*

INTRODUCTION

The ever-increasing complexity of architectural projects and the rapid advancement of 3D printing technologies have increased the demand for a more adaptive and flexible approach to assembly strategies. Current Design for Assembly (DfA) methods often fall short in addressing unpredictable material behavior and unknown parameters, particularly in the context of 3D printed structures. In response to this challenge, this paper introduces a novel workflow that embraces change and adapts to variations in tolerances and geometry during the assembly process. By integrating techniques such as 3D scanning, laser project and early 1:1 prototyping, we can build an assembly model early on in the design process that is both adaptive and responsive to changes in structural and site conditions.

BACKGROUND

DfA methods have been widely used in the construction and manufacturing industries to streamline assembly processes, optimize the use of materials, and minimize production costs. However, these methods have inherent limitations when dealing with unexpected material behavior, particularly in the context of complex architectural projects. The primary challenge posed by unexpected material behavior in traditional DfA methods stems from their reliance on precise and predetermined design parameters. DfA methods typically assume that materials will behave consistently and within established tolerances throughout the assembly process. However, in reality, materials can exhibit unpredictable behavior due to various factors such as manufacturing inconsistencies, environmental conditions, and handling or transportation-related stresses.

This unpredictability can lead to discrepancies between the planned assembly process and the actual on-site conditions, resulting in time-consuming adjustments, increased labor costs, and potentially compromising the structural integrity or aesthetics of the final product. Furthermore, when DfA methods are applied to novel materials or unconventional fabrication processes, the lack of historical data and established tolerances can exacerbate these challenges, making it difficult to anticipate and mitigate potential issues.

One way to manage varying material behavior is to utilize Machine Learning (ML) models for predicting material behavior, as suggested by Rossi et al. (2022). However, this approach requires a comprehensive understanding of the specific material properties and are often limited to a specific type or mix of material. Therefore, it can't be generalized to other materials without the need to either recompose a dataset with the new material variation or comparing the new material behavior to the trained baseline.

Several researchers have looked into various techniques to tackle the issues present in the assembly of these kind of structures. Generally, the focus lies on either the connection components or the assembly sequence itself. Concerning connection elements, one strategy is the use of adaptive component design, which involves the creation of parts with inherent flexibility to accommodate tolerances during assembly (Moreira et al., 2017). Additionally, some studies have concentrated on optimizing the assembly sequence to minimize the effects of uncertainties. Techniques such as genetic algorithms and physical simulations can assist in pinpointing optimal sequences (Kao et al. 2017; Wang et al., 2021). By generating a multitude of variations and pre-emptively formulating alternative plans, potential assembly problems can be addressed more effectively. Another approach involves employing monitoring systems (Jiang et al., 2022), with digital twins utilized during the process to surmount DfA limitations by delivering real-time visualizations of the physical assembly. This continuous monitoring allows for immediate simulations and optimization.

Furthermore, digital fabrication and assembly methods have been gaining traction in the recent years, in particular within the realm of robotic fabrication applied to timber structure (Gharbia et al. 2020; Kunic et al. 2021). By combining the flexibility of robotic arms, sensor data and real-time feedback loops, it becomes possible to make adjustments during the fabrication and assembly process (Rogeau et al. 2020; Ye et al. 2022).

METHODOLOGY

This work builds on the top of existing research on Assembly Information Modeling (AIM) (Lharchi et al. 2021), where the goal is to build a detailed assembly model that integrates all the necessary information (3D geometry, assembly instructions, tolerances, order, etc.).

Figure 1
Overview of the methodology

The framework we developed for this purpose (figure 1) is divided into two primary stages: pre-production:

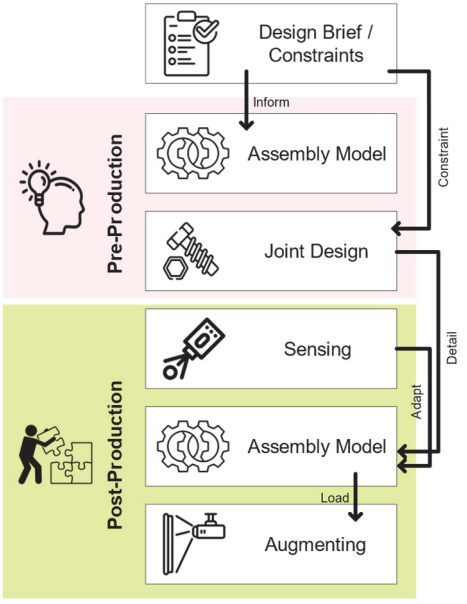
1. Pre-Production:
   a. Initial Model Creation: The first step involves the creation of an initial assembly model using available boundary conditions (CAD models or 3D scans) and initial design sketches. Afterwards, we determine tolerances and potential needs of adjustments.
   b. Joint Design: Based on the identified tolerances, we design a joint that can accommodate variations in geometry and material properties.

2. Post-Production:
   a. Sensing and Adapting: After the production, we perform a 3D scan to identify any shrinkage and material changes that may have occurred during the production or drying process. We update the assembly model to account for any deviations observed in the 3D scans.
   b. Augmenting the Assembly: The final stage involves the on-site assembly of the structure. To ensure accurate placement and alignment, we employ laser projection technology that loads information directly from the adapted assembly model.

CASE STUDY: LARGE-SCALE DEMONSTRATOR

To demonstrate the efficacity of the workflow described here, we present a case study featuring a built demonstrator of 7x5m, exhibited at the Aedes in Berlin in 2022/23. The project "*Radicant*" is a bespoke wall-paneling system made out from a biopolymer composites. Overall, 24 panels of variable sizes were produced using robotic 3d printing.

The workflow described above was used in the assembly design of a large-scale demonstrator.



Pre-Production: Initial Assembly Model

The first step in our methodology involves the development of an initial assembly model, which serves as a starting point for the entire assembly process. At this stage, a Low Level of Detail (LOD) geometry is generated to represent the structure's basic form. This model is purposely designed to be agnostic to connections, allowing for greater flexibility in the later stages of the process.

In order to negotiate between the position of the attachments to the wall, existing elements, and material requirements, we used an agent-based algorithm to satisfy most of our constraints. The position was adjusted incrementally until it converges to a stable solution (figure 2). Panels were assigned two or three connection points (depending on their size), and those points were considered as free agents with a set of rules to satisfy the constraints.

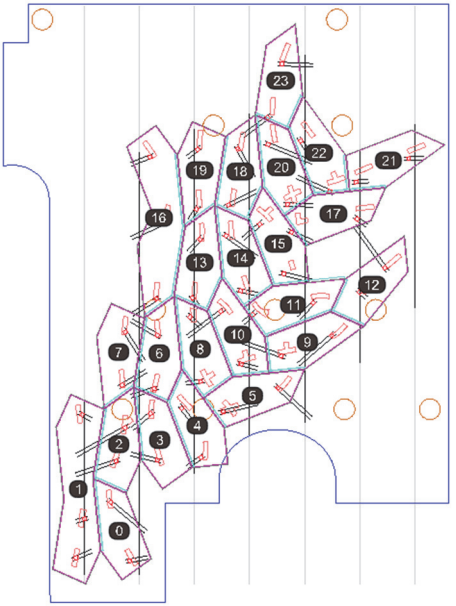


Figure 2
Structure Overview (with branches and beams)

After that, we proceeded to generate possible assembly sequences. By using the LOD model as a reference, we were able to visualize the process and determine the required equipment for each step. For the generation, we used an iterative approach that systematically explored various combinations of component assembly order. Each potential sequence was then evaluated based on factors such as ease of assembly, required labor and equipment, and timeline.

Pre-Production: Joint Design

The joint design phase of our project was a critical stage where we carefully evaluated and considered various options to find the most effective and practical solution for assembling our panels. Through a thorough exploration process, we finally decided on a sandwich joint approach that involved three main components.

The first component was a timber piece embedded within the printed model during the fabrication process. This provided a solid and reliable connection point, as we knew that the biopolymer material, we were using was unsuitable for direct drilling or screwing. This piece acted as the foundation of the joint design, providing stability and strength to the overall structure.

The second component was a second piece attached to the panel, which mirrored the first one. This was designed to create a secure connection between the two components, ensuring that they were held together tightly and could withstand any external forces. This component was also critical in ensuring that the structure remained stable and could withstand any stress during assembly and throughout its lifespan.

Finally, we used a profile to offset the panel from the wall, which helped prevent direct contact between the panel and the wall. This minimized the risk of damage to the panel, especially during any movement or vibrations, and ensured the overall stability of the structure.

During the joint design process, we paid particular attention to the User Manipulation Zone (UMZ) and the reachability of the fasteners. Since we had a structure that was fragile and restricted reachability and freedom of movement during the assembly (on a scaffolding, 7m above the ground), we had to ensure that the joint design was fool proof for the on-site team.

Post-Production: Sensing and Adapting

Following the drying process (after 3d printing), we moved on to the sensing phase of our methodology. In this stage, we employed an industrial Faro scanner to perform 3D scans of the final panels. This allowed us to accurately capture the final geometry of the panels, including any changes or deformations that may have occurred during the drying process.

Once the 3D scans were complete, we digitized the position of the backplate on each panel, creating a precise representation of the real-world conditions. This digitized information was then loaded back into the assembly model, updating it

Figure 3
3D scanned data overlay with the initial assembly model

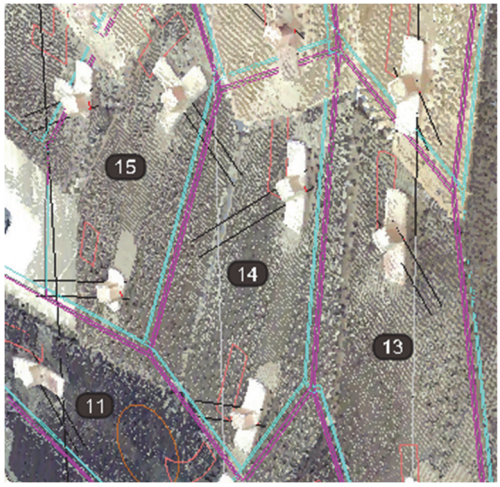


Figure 4
Assembly augmentation using laser projection



with the detailed geometry of the final panels (figure 3). By integrating the scanned data into the model, we were able to ensure that the assembly model accurately represented the actual physical components.

With the updated assembly model in place, we proceeded to verify if the tolerances were still respected in light of the changes that occurred during the drying process. This involved examining the fit between components, analyzing load distribution, and ensuring that the connections maintained their structural integrity.

In case any discrepancies were found, we ran the assembly computation again, making necessary adjustments to the model to account for the observed deviations. This iterative process allowed us to refine and optimize the assembly model, ensuring that it remained adaptive and responsive to changes in material behavior and site conditions.

Post-Production: Augmenting the assembly

On-site, we used a LAP laser projection system to assist the on-site assembly team. The assembly model was flattened into 2d and then exported in the native format of the project (DXF). The projection provided real-time guidance (figure 4) on the specific order of assembly and played a crucial role in accurately identifying the exact positions of the elements on the wall. The system provided information on where to drill, vertical and horizontal alignment, and the correct spacing between the components, thus reducing greatly the likelihood of errors.

The continuous monitoring and validation process ensured that any deviations or misalignments were promptly identified and corrected.

RESULTS AND DISCUSSION

The implementation of the proposed workflow in our large-scale demonstrator project yielded significant improvements in the assembly process and overall project outcomes. By integrating various techniques such as agent based algorithms, 3D scanning, and laser projection, we were able to develop an adaptive and responsive assembly model that effectively addressed unpredictable material behavior and varying site conditions (figure 5).

The sandwich joint approach proved to be an effective solution for connecting the panels, accommodating the unique material properties, and

ensuring a robust and reliable connection between the components. The use of an agent-based algorithm for optimizing attachment positions allowed for a more efficient assembly process and reduced the likelihood of issues during construction.

The sensing phase, which involved 3D scanning of the final panels and updating the assembly model, ensured a high level of accuracy and adaptability throughout the construction process. By continuously monitoring and adjusting the assembly model, we were able to minimize potential discrepancies and maintain the overall quality and integrity of the structure.

The augmenting the assembly phase, which utilized a laser projection system to guide the on-site assembly team, significantly streamlined the construction process and enhanced the overall accuracy of the final product. The laser projection system enabled precise placement of the elements on the wall and provided real-time guidance on the specific order of assembly, resulting in a more efficient and cost-effective project outcome.

However, it is important to acknowledge some limitations of the proposed workflow. One limitation is the dependency on advanced technologies like industrial 3D scanners and laser projection systems, which might not be readily accessible or affordable for all projects.

Additionally, the workflow may require a significant investment of time and resources in the initial stages, particularly for model creation and joint design. Lastly, while the workflow has been demonstrated to be effective in the context of 3D printed structures, further research and testing are needed to evaluate its adaptability to other construction materials and methods.

Despite its limitations, the novel workflow presented in this paper represents a significant advancement in the field of DfA methods, offering a more adaptive and responsive solution for complex architectural projects. By building upon the insights and successes of this study, future research can continue to refine and expand upon this approach,



Figure 5
Final Installation

ultimately contributing to the development of more efficient and sustainable construction practices.

CONCLUSION

In this paper, we presented a novel workflow for addressing the challenges associated with the assembly of complex architectural projects, particularly in the context of 3D printed structures. By integrating advanced techniques such as 3D scanning, laser projection, and early 1:1 prototyping, we demonstrated the development of an adaptive and responsive assembly model that effectively accommodates unpredictable material behavior and varying site conditions.

The successful implementation of this workflow in our large-scale demonstrator project showcases the potential for its widespread application in future complex architectural projects. By adopting such an adaptive and flexible approach to assembly strategies, architects and construction professionals can optimize the use of materials, minimize production costs, and streamline assembly processes, ultimately leading to more efficient and cost-effective project outcomes.

AUTHORS CONTRIBUTION

The manuscript was written with the contribution of all authors. All authors have approved the final version of the manuscript. Lharchi, A., design concept, computational modeling framework development, 3d print strategy, fabrication, installation, writing – original draft, reviewing and editing, Tamke, M., project conceptualization, funding acquisition, methodology, writing – original draft, reviewing and editing, installation, supervision, funding acquisition, Nicholas, P. project conceptualization, methodology, design concept, writing – original draft, reviewing and editing, supervision, Valipour Goudarzi H., design concepts, visualization, prototyping, fabrication, installation, Eppinger, C., design concept, 3D print hardware development, prototyping, fabrication, installation, Sonne, K., prototyping, fabrication, installation, Rossi, G., material specification strategy, Ramsgaard Thomsen, M. project conceptualization, methodology, design concept, writing – review and editing, lab infrastructure, supervision, funding acquisition.

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